

ZEZHONG (ZEDDY) DING

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RESEARCH INTERESTS

My research focuses on data management, systems, and artificial intelligence over big data and large models, in particular, big graphs and foundation models.

EDUCATION

University of Science and Technology of China (USTC) 2022.09 – Present
Ph.D. in Computer Science and Technology (successive master-doctor program)
Thesis: Graph Partitioning and Training Optimization in Graph Computing Systems, supervised by Prof. Xike Xie

Hong Kong Baptist University (HKBU) 2025.11 – 2026.02
Visiting Student at Department of Computer Science
Research topic: Dynamic Graph Systems, supervised by Prof. Jianliang Xu

Ocean University of China (OUC) 2018.09 – 2022.06
B.Eng. (Honors) in Computer Science and Technology (GPA: 97.72/100, Class Rank: 1/234)
GPA: 97.72/100, Class Rank: 1/234. Awarded Outstanding Thesis and Honors Degree.

PUBLICATIONS (* indicates equal contribution)

- [1] [SIGMOD-24] **Zezhong Ding**, Yongan Xiang, Shangyou Wang, Xike Xie, and S. K. Zhou. “**Play like a Vertex: A Stackelberg Game Approach for Streaming Graph Partitioning.**” In *Proc. ACM Manag. Data*, 2024.
- [2] [TC-25] **Zezhong Ding**, Deyu Kong, Zhuoxu Zhang, Xike Xie and Jianliang Xu. “**ClusPar: A Game-Theoretic Approach for Efficient and Scalable Streaming Edge Partitioning.**” In *IEEE Trans. Computers*, 2025.
- [3] [SIGMOD-25] Yongan Xiang*, **Zezhong Ding***, Rui Guo, Shangyou Wang, Xike Xie and S. K. Zhou. “**Capsule: An Out-of-Core Training Mechanism for Colossal GNNs.**” In *Proc. ACM Manag. Data*, 2025.
- [4] [NeurIPS-25] Jin Li*, **Zezhong Ding*** and Xike Xie. “**DuetGraph: Coarse-to-Fine Knowledge Graph Reasoning with Dual-Pathway Global-Local Fusion.**” In *Advances in Neural Information Processing Systems*, 2025.
- [5] [ACL-25] Yukun Cao, Shuo Han, Zengyi Gao, **Zezhong Ding**, Xike Xie, S. K. Zhou. “**GraphInsight: Unlocking Insights in Large Language Models for Graph Structure Understanding.**” In *Proc. ACL*, 2025.
- [6] [SIGMOD-26] Rui Guo*, **Zezhong Ding***, Xike Xie, Jianliang Xu. “**SWIFT: Enabling Large-Scale Temporal Graph Learning on a Single Machine.**” In *Proc. ACM Manag. Data*, 2025.
- [7] [KDD-26] Huhao Guan*, **Zezhong Ding***, Ao Ke, Xike Xie, S. K. Zhou. “**SIGHP: Scalable Information-Guided Hypergraph Partitioner.**” In *Proc. Knowledge Discovery and Data Mining*, 2026.
- [8] [KDD-26] Shangyou Wang*, **Zezhong Ding***, Xike Xie, “**SamGoG: A Sampling-Based Graph-of-Graphs Framework for Imbalanced Graph Classification.**” In *Proc. Knowledge Discovery and Data Mining*, 2026.
- [9] [ICML-26] Haokun Liu*, **Zezhong Ding***, Xike Xie, “**Learning Graph Foundation Models on Riemannian Graph-of-Graphs.**” In *Proc. International Conference on Machine Learning*, 2026.
- [10] [SC-26] **Zezhong Ding**, Rui Guo, Wenbo Zhen, Junlin Lv, Jianliang Xu, Xike Xie. “**DOLPHIN: Scalable Disk-RAM-GPU Pipelined Training for Massive Temporal GNNs.**” In *Proc. High Performance Computing, Networking, Storage, and Analysis*, 2026. **Best Student Paper Finalist**

HONORS AND AWARDS

- SC Best Student Paper Finalist, SC 2026
- Finalist for Grand Prize (Top 1%), *International MCM/ICM (USA)* 2021
- Gold Medal, *International Genetically Engineered Machine Competition (USA)* 2021
- AEON Scholarship, *Aeon (Japan)* 2020

ACADEMIC SERVICE

- **Reviewer:** NeurIPS 2026, ICML 2026, SIGMOD 2025 (ARI), TKDD 2024
- **External Reviewer:** VLDB(J) 2026, NeurIPS 2024, ICDE 2024-2026, SIGIR 2022

TALKS AND PRESENTATIONS

- Graph Partitioning: S5P, Oral Presentation [Slides], SIGMOD 2025, Berlin, Germany 2025.06
- GNN System: Capsule, Oral Presentation [Slides], SIGMOD 2024, Santiago, Chile 2024.06

TEACHING EXPERIENCE

Undergraduate Teaching Assistant in OUC 2020 – 2021

- **Core CS Courses:** Data Structures and Algorithms, Introduction to Computer Systems, Computer Network, Object-Oriented Programming, High-level Programming Language, and Numerical Analysis.